

# **Assignment #1 : Computer and Network Security**

## **Part 1: Executive Summary**

An investigation into the provided .pcap file has found evidence of suspicious activity in the network through repeated attempts to make unauthorized connections into the network using various protocols. Current security measures should be maintained and monitored and the origins of these attempts should be investigated further.

## **Part 2: Environment and Scope**

This section will clarify the environment and scope for this analysis. The investigation will be conducted using a provided .pcap file with network traffic collected on an internal network. The analysis will be conducted using Wireshark to filter and decode the traffic and draw conclusions.

## **Part 3: Hosts Identified**

This section will identify major host IP addresses and MAC addresses present in the packet capture.

When sorting the IPs with the highest amount of traffic, the following significant IPs were found:

10.10.30.10 – This IP's traffic consisted of various HTTP/TCP requests sent from it originating at port 80 indicating that this is a web server, but at times it also works as a client querying for HTTP pages. This is a private IP address. Corresponding MAC address: 02:33:0a:0a:1e:0a.

198.51.100.20 – This IP's traffic consisted of various HTTP/TCP requests originating at port 80 indicating that this is a web server. This is a public IP address.

10.10.20.25 – This IP address looks to be a client querying to different webpages, a TLS connection, and DNS requests.

10.10.20.53 – This IP appears to be a client that is querying DNS repeatedly.

## Part 4: Protocol Analysis

This section will identify and analyze the protocols found in the packet capture.

Major protocols present: TCP, UDP, HTTP, DNS, ICMP, ARP, SSH

### TCP

Filter: tcp

|    |          |             |             |     |    |                                                   |
|----|----------|-------------|-------------|-----|----|---------------------------------------------------|
| 11 | 2.080000 | 10.10.20.60 | 10.10.30.10 | TCP | 54 | 59333 → 80 [SYN] Seq=0 Win=64240 Len=0            |
| 12 | 2.086000 | 10.10.30.10 | 10.10.20.60 | TCP | 54 | 80 → 59333 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 |
| 13 | 2.092000 | 10.10.20.60 | 10.10.30.10 | TCP | 54 | 59333 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0      |

Figure 1. Three TCP packets demonstrating a complete 3-way-handshake.

Explanation: The first [SYN] is sent from the client (10.10.20.60) to the server (10.10.30.10) asking to connect or synchronize. The server responds with [SYN, ACK], the [SYN] to set the starting sequence number of the packets, and the [ACK] acknowledges/confirms the message is received. The final [ACK] confirms the message is received by the client and starts the connection.

### SSH

Filter: ssh

| No.   | Time       | Source      | Destination | Protocol | Length | Info                                          |
|-------|------------|-------------|-------------|----------|--------|-----------------------------------------------|
| 2493  | 45.020000  | 10.10.30.10 | 10.10.20.61 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 2497  | 45.040000  | 10.10.20.61 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 4286  | 75.020000  | 10.10.30.10 | 10.10.20.63 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 4288  | 75.040000  | 10.10.20.63 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 11578 | 105.020000 | 10.10.30.10 | 10.10.20.65 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 11583 | 105.040000 | 10.10.20.65 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 13374 | 135.020000 | 10.10.30.10 | 10.10.20.67 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 13376 | 135.040000 | 10.10.20.67 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 15167 | 165.020000 | 10.10.30.10 | 10.10.20.69 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 15170 | 165.040000 | 10.10.20.69 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 16980 | 195.020000 | 10.10.30.10 | 10.10.20.71 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 16982 | 195.040000 | 10.10.20.71 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 18771 | 225.020000 | 10.10.30.10 | 10.10.20.73 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 18775 | 225.040000 | 10.10.20.73 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 20567 | 255.020000 | 10.10.30.10 | 10.10.20.75 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 20571 | 255.040000 | 10.10.20.75 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 22362 | 285.020000 | 10.10.30.10 | 10.10.20.77 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 22364 | 285.040000 | 10.10.20.77 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 24171 | 315.020000 | 10.10.30.10 | 10.10.20.79 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 24175 | 315.040000 | 10.10.20.79 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 25966 | 345.020000 | 10.10.30.10 | 10.10.20.81 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 25968 | 345.040000 | 10.10.20.81 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 27757 | 375.020000 | 10.10.30.10 | 10.10.20.83 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 27760 | 375.040000 | 10.10.20.83 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 29557 | 405.020000 | 10.10.30.10 | 10.10.20.85 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 29559 | 405.040000 | 10.10.20.85 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 31348 | 435.020000 | 10.10.30.10 | 10.10.20.87 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 31350 | 435.040000 | 10.10.20.87 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |
| 33148 | 465.020000 | 10.10.30.10 | 10.10.20.89 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)      |
| 33152 | 465.040000 | 10.10.20.89 | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0) |

|                                                                                                                                         |      |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|
| Frame 2497: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits)                                                           | 0000 | 02 33 0a 0a 14 01 |
| Ethernet II, Src: MS-NLB-PhysServer-32_13:0a:0a:14:3d (02:33:0a:0a:14:3d), Dst: MS-NLB-PhysServer-32_13:0a:0a:14:01 (02:33:0a:0a:14:01) | 0010 | 00 44 e9 18 40 00 |
| Internet Protocol Version 4, Src: 10.10.20.61, Dst: 10.10.30.10                                                                         | 0020 | 1e 0a e4 98 00 16 |
| Transmission Control Protocol, Src Port: 58520, Dst Port: 22, Seq: 1, Ack: 24, Len: 28                                                  | 0030 | fa f0 05 ad 00 00 |
| SSH Protocol                                                                                                                            | 0040 | 61 69 6e 69 6e 67 |
| Protocol: SSH-2.0-TrainingClient_1.0                                                                                                    | 0050 | 0d 0a             |
| [Direction: Client to Server]                                                                                                           |      |                   |

Figure 2. Repeated attempts at an SSH connection by various clients.

Explanation: There are repeated ssh connection attempts from various source IPs to the same destination IP, 10.10.30.10. These attempts happen 30 seconds apart from each other, and receive a response 0.02 seconds later, signaling that the connection likely failed to occur. Due to the repeated attempts from various IPs to a single target, over consistent timeframes, that ultimately failed, this traffic seems to be suspicious as though an individual is repeatedly attempting to connect over SSH to the target.

## HTTP

Filter: http.response.code > 400

|       |             |               |               |          |                                                         |
|-------|-------------|---------------|---------------|----------|---------------------------------------------------------|
| 77157 | 1200.050000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77270 | 1201.800000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77394 | 1203.550000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77508 | 1205.300000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77618 | 1207.050000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77732 | 1208.800000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77847 | 1210.550000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 77970 | 1212.300000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78082 | 1214.050000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78202 | 1215.800000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78309 | 1217.550000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78434 | 1219.300000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78557 | 1221.050000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78664 | 1222.800000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78760 | 1224.330000 | 10.10.30.10   | 10.10.20.89   | HTTP     | 153 HTTP/1.1 404 Not Found (text/plain)                 |
| 78773 | 1224.550000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 78882 | 1226.300000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 79010 | 1228.050000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 79128 | 1229.800000 | 10.10.30.10   | 198.51.100.77 | HTTP/1.1 | 177 HTTP/1.1 401 Unauthorized , JSON (application/json) |
| 81269 | 1265.070000 | 198.51.100.20 | 10.10.20.25   | HTTP     | 153 HTTP/1.1 404 Not Found (text/plain)                 |

```

HTTP/1.1 401 Unauthorized\r\n
Content-Type: application/json\r\n
Content-Length: 23\r\n
Connection: close\r\n
\r\n
[Request in frame: 77153]
[Time since request: 30.000000 milliseconds]
[Request URI: /admin/login]
[Full request URI: http://intranet.hatter.lab/admin/login]
File Data: 23 bytes
JavaScript Object Notation: application/json

```

Figure 3. Repeated attempts at querying unauthorized HTTP data.

There are multiple HTTP requests to Unauthorized webpages, which, looking at the packet details, show it is to the URL <http://intranet.hatter.lab/admin/login>. This URL seems to be for an admin panel, and the individual sending the requests does not have access to it. Due to the large number of Unauthorized requests to this URL in a short amount of time in the time column, it may be interpreted as someone attempting various credentials to gain unauthorized access to the admin panel.

Filter: `http.response.code==200`

|        |             |               |             |      |     |          |     |    |              |
|--------|-------------|---------------|-------------|------|-----|----------|-----|----|--------------|
| 107123 | 1696.830000 | 198.51.100.20 | 10.10.20.96 | HTTP | 508 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107145 | 1697.250000 | 10.10.30.10   | 10.10.20.97 | HTTP | 538 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107154 | 1697.290000 | 10.10.30.10   | 10.10.20.15 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107170 | 1697.670000 | 10.10.30.10   | 10.10.20.98 | HTTP | 478 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107200 | 1698.080000 | 10.10.30.10   | 10.10.20.69 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107202 | 1698.090000 | 10.10.30.10   | 10.10.20.99 | HTTP | 703 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107219 | 1698.510000 | 10.10.30.10   | 10.10.20.25 | HTTP | 538 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107242 | 1698.870000 | 198.51.100.20 | 10.10.20.80 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107249 | 1698.930000 | 198.51.100.20 | 10.10.20.15 | HTTP | 613 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107268 | 1699.350000 | 10.10.30.10   | 10.10.20.40 | HTTP | 748 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107291 | 1699.660000 | 10.10.30.10   | 10.10.20.91 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107302 | 1699.770000 | 198.51.100.20 | 10.10.20.60 | HTTP | 237 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107315 | 1700.190000 | 10.10.30.10   | 10.10.20.61 | HTTP | 868 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107338 | 1700.450000 | 10.10.30.10   | 10.10.20.40 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107351 | 1700.610000 | 198.51.100.20 | 10.10.20.62 | HTTP | 237 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107366 | 1701.030000 | 10.10.30.10   | 10.10.20.63 | HTTP | 718 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107389 | 1701.240000 | 10.10.30.10   | 10.10.20.70 | HTTP | 937 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107400 | 1701.450000 | 198.51.100.20 | 10.10.20.64 | HTTP | 463 | HTTP/1.1 | 200 | OK | (text/plain) |
| 107415 | 1701.870000 | 10.10.30.10   | 10.10.20.65 | HTTP | 328 | HTTP/1.1 | 200 | OK | (text/plain) |

```

Content-Type: text/plain\r\n
Content-Length: 369\r\n
Connection: close\r\n
\r\n
[Request in frame: 107121]
[Time since request: 30.000000 milliseconds]
[Request URI: /docs/index.html]
[Full request URI: http://portal.example.com/docs/index.html]
File Data: 369 bytes
Line-based text data: text/plain (1 lines)
[... Training response 4035: sample content sample content sample content sample content sample content sample content

```

Figure 4. Plaintext HTTP responses with sample text.

There are also a good number of plain text HTTP traffic with the 200 OK response. Investigating the packet details on any of these show either “sample content” or “This is synthetic classroom network traffic” to be in the payload data. These packets are un concerning and reflect the fact they are in a simulated classroom environment.

## DNS

Filter: dns.flags.rcode==3

|    |                         |        |                                       |
|----|-------------------------|--------|---------------------------------------|
| 80 | Standard query response | 0x390c | No such name A missing-0.hatter.lab   |
| 81 | Standard query response | 0xd9d8 | No such name A missing-17.hatter.lab  |
| 81 | Standard query response | 0xa413 | No such name A missing-34.hatter.lab  |
| 81 | Standard query response | 0x9238 | No such name A missing-51.hatter.lab  |
| 81 | Standard query response | 0x8fc2 | No such name A missing-68.hatter.lab  |
| 81 | Standard query response | 0x801e | No such name A missing-85.hatter.lab  |
| 82 | Standard query response | 0x6c06 | No such name A missing-102.hatter.lab |
| 82 | Standard query response | 0xccbb | No such name A missing-119.hatter.lab |
| 82 | Standard query response | 0x918e | No such name A missing-136.hatter.lab |

Figure 5. Repeated DNS requests to nonexistent subdomains.

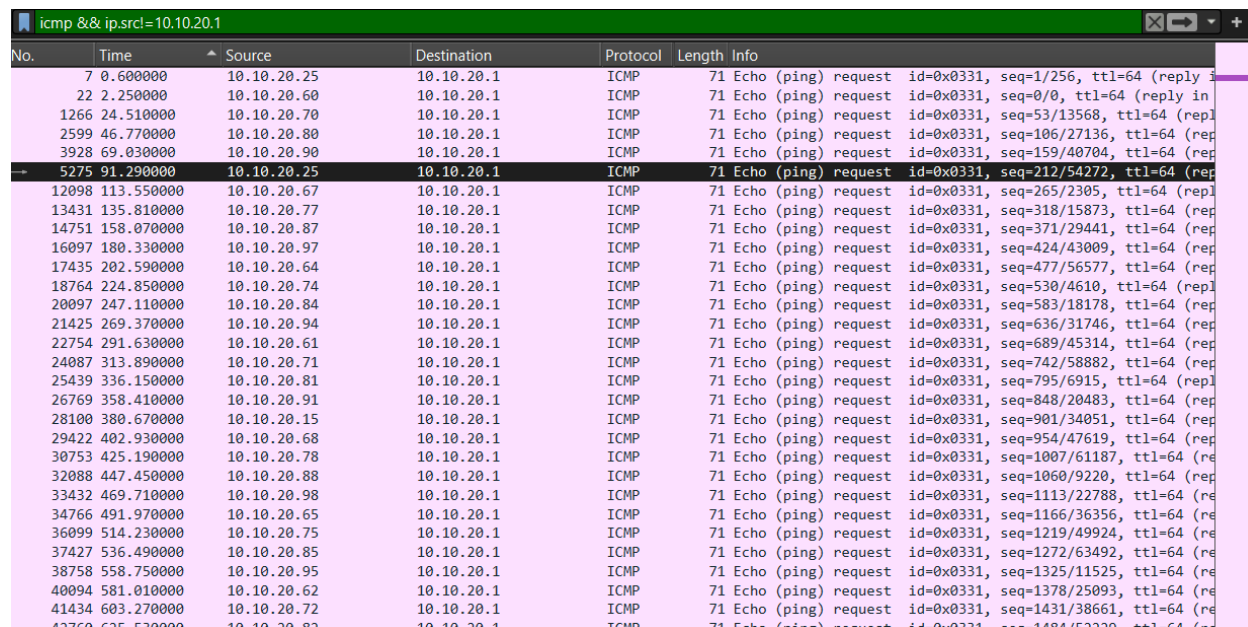
Most DNS traffic in this file is querying to the same few domains repeatedly, those being: intranet.hatter.lab, updates.hatter.lab, portal.example.com, status.hatter.lab, docs.hatter.lab, [www.example.org](http://www.example.org). There are several queries that get a “No such

name” response from the server attempting to query domains with the format missing-x.hatter.lab.

This response indicates that someone is attempting to reach multiple different domains with various variables for x. However, none of them exist, so the DNS response replies that there is “No such name.” Due to the volume and frequent timing of the queries, this may be evidence of a subdomain brute force attack against the domain hatter.lab. The attacker is attempting to find hidden subdomains by querying random URLs.

## ICMP

Filter: icmp && ip.src!=10.10.20.1



| No.    | Time       | Source      | Destination | Protocol | Length | Info                                                                  |
|--------|------------|-------------|-------------|----------|--------|-----------------------------------------------------------------------|
| 7      | 0.600000   | 10.10.20.25 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1/256, ttl=64 (reply in 64ms)      |
| 22     | 2.250000   | 10.10.20.60 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=0/0, ttl=64 (reply in 64ms)        |
| 1266   | 24.510000  | 10.10.20.70 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=53/13568, ttl=64 (reply in 64ms)   |
| 2599   | 46.770000  | 10.10.20.80 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=106/27136, ttl=64 (reply in 64ms)  |
| 3928   | 69.030000  | 10.10.20.90 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=159/40704, ttl=64 (reply in 64ms)  |
| → 5275 | 91.290000  | 10.10.20.25 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=212/54272, ttl=64 (reply in 64ms)  |
| 12098  | 113.550000 | 10.10.20.67 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=265/2305, ttl=64 (reply in 64ms)   |
| 13431  | 135.810000 | 10.10.20.77 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=318/15873, ttl=64 (reply in 64ms)  |
| 14751  | 158.070000 | 10.10.20.87 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=371/29441, ttl=64 (reply in 64ms)  |
| 16097  | 180.330000 | 10.10.20.97 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=424/43009, ttl=64 (reply in 64ms)  |
| 17435  | 202.590000 | 10.10.20.64 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=477/56577, ttl=64 (reply in 64ms)  |
| 18764  | 224.850000 | 10.10.20.74 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=530/4610, ttl=64 (reply in 64ms)   |
| 20097  | 247.110000 | 10.10.20.84 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=583/18178, ttl=64 (reply in 64ms)  |
| 21425  | 269.370000 | 10.10.20.94 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=636/31746, ttl=64 (reply in 64ms)  |
| 22754  | 291.630000 | 10.10.20.61 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=689/45314, ttl=64 (reply in 64ms)  |
| 24087  | 313.890000 | 10.10.20.71 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=742/58882, ttl=64 (reply in 64ms)  |
| 25439  | 336.150000 | 10.10.20.81 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=795/6915, ttl=64 (reply in 64ms)   |
| 26769  | 358.410000 | 10.10.20.91 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=848/20483, ttl=64 (reply in 64ms)  |
| 28100  | 380.670000 | 10.10.20.15 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=901/34051, ttl=64 (reply in 64ms)  |
| 29422  | 402.930000 | 10.10.20.68 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=954/47619, ttl=64 (reply in 64ms)  |
| 30753  | 425.190000 | 10.10.20.78 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1007/61187, ttl=64 (reply in 64ms) |
| 32088  | 447.450000 | 10.10.20.88 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1060/9220, ttl=64 (reply in 64ms)  |
| 33432  | 469.710000 | 10.10.20.98 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1113/22788, ttl=64 (reply in 64ms) |
| 34766  | 491.970000 | 10.10.20.65 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1166/36356, ttl=64 (reply in 64ms) |
| 36099  | 514.230000 | 10.10.20.75 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1219/49924, ttl=64 (reply in 64ms) |
| 37427  | 536.490000 | 10.10.20.85 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1272/63492, ttl=64 (reply in 64ms) |
| 38758  | 558.750000 | 10.10.20.95 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1325/11525, ttl=64 (reply in 64ms) |
| 40094  | 581.010000 | 10.10.20.62 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1378/25093, ttl=64 (reply in 64ms) |
| 41434  | 603.270000 | 10.10.20.72 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1431/38661, ttl=64 (reply in 64ms) |
| 42760  | 625.530000 | 10.10.20.82 | 10.10.20.1  | ICMP     | 71     | Echo (ping) request id=0x0331, seq=1484/52228, ttl=64 (reply in 64ms) |

Figure 6. Pings from various IPs every 20 seconds to a single destination.

There are ICMP packets from various IP addresses aiming towards one target destination IP, 10.10.20.1. They are all sent approximately 20 seconds apart. Various devices test connectivity to a single device. Due to the steady timeframe, it is likely that it is normal activity and is not currently proof of malicious activity.

## ARP

Filter: arp

| No.   | Time       | Source                  | Destination             | Protocol | Length | Info                                  |
|-------|------------|-------------------------|-------------------------|----------|--------|---------------------------------------|
| 1     | 0.000000   | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.25  |
| 2     | 0.010000   | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 3     | 0.100000   | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.53? Tell 10.10.20.25 |
| 4     | 0.110000   | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.53 is at 02:33:0a:0a:14:35   |
| 24    | 2.300000   | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.60  |
| 25    | 2.310000   | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 5245  | 90.920000  | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.99  |
| 5246  | 90.930000  | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 16039 | 179.540000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.95  |
| 16040 | 179.550000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 21352 | 268.160000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.91  |
| 21355 | 268.170000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 26671 | 356.780000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.87  |
| 26672 | 356.790000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 31968 | 445.400000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.83  |
| 31969 | 445.410000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 37280 | 534.020000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.79  |
| 37281 | 534.030000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 42579 | 622.640000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.75  |
| 42580 | 622.650000 | MS-NLB-PhysServer-32... | MS-NLB-PhysServer-32... | ARP      | 42     | 10.10.20.1 is at 02:33:0a:0a:14:01    |
| 47876 | 711.260000 | MS-NLB-PhysServer-32... | Broadcast               | ARP      | 42     | Who has 10.10.20.1? Tell 10.10.20.71  |

Figure 7. Periodic ARP requests from MS-NLB-PhysServer-32\_13.

The ARP requests show a mixture of broadcast messages and responses. The requests are made by different IP addresses following the 10.10.20.x format. They are all querying for the MAC address of the same IP, 10.10.20.1, at regular time intervals apart, approximately 88 seconds. However, this is not enough to conclude malicious activity, but it should be investigated further.

## Part 5: Timeline of Relevant Events

| No.   | Time       | Source        | Destination | Protocol | Length | Info                                                                 |
|-------|------------|---------------|-------------|----------|--------|----------------------------------------------------------------------|
| 16    | 2.130000   | 10.10.30.10   | 10.10.20.60 | HTTP     | 153    | HTTP/1.1 404 Not Found (text/plain)                                  |
| 73    | 4.020000   | 10.10.20.53   | 10.10.20.60 | DNS      | 80     | Standard query response 0x390c No such name A missing-0.hatter.lab   |
| 480   | 11.330000  | 10.10.20.53   | 10.10.20.93 | DNS      | 81     | Standard query response 0xd9d8 No such name A missing-17.hatter.lab  |
| 928   | 18.640000  | 10.10.20.53   | 10.10.20.83 | DNS      | 81     | Standard query response 0xa413 No such name A missing-34.hatter.lab  |
| 1353  | 25.950000  | 10.10.20.53   | 10.10.20.73 | DNS      | 81     | Standard query response 0x9238 No such name A missing-51.hatter.lab  |
| 1792  | 33.260000  | 10.10.20.53   | 10.10.20.63 | DNS      | 81     | Standard query response 0x8fc2 No such name A missing-68.hatter.lab  |
| 2219  | 40.570000  | 10.10.20.53   | 10.10.20.96 | DNS      | 81     | Standard query response 0x801e No such name A missing-85.hatter.lab  |
| 2354  | 42.870000  | 198.51.100.20 | 10.10.20.71 | HTTP     | 153    | HTTP/1.1 404 Not Found (text/plain)                                  |
| 2493  | 45.020000  | 10.10.30.10   | 10.10.20.61 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)                             |
| 2497  | 45.040000  | 10.10.20.61   | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0)                        |
| 2676  | 47.880000  | 10.10.20.53   | 10.10.20.86 | DNS      | 82     | Standard query response 0x6c06 No such name A missing-102.hatter.lab |
| 3107  | 55.190000  | 10.10.20.53   | 10.10.20.76 | DNS      | 82     | Standard query response 0xccbb No such name A missing-119.hatter.lab |
| 3536  | 62.500000  | 10.10.20.53   | 10.10.20.66 | DNS      | 82     | Standard query response 0x918e No such name A missing-136.hatter.lab |
| 3976  | 69.810000  | 10.10.20.53   | 10.10.20.99 | DNS      | 82     | Standard query response 0x98ce No such name A missing-153.hatter.lab |
| 4286  | 75.020000  | 10.10.30.10   | 10.10.20.63 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)                             |
| 4288  | 75.040000  | 10.10.20.63   | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0)                        |
| 4425  | 77.120000  | 10.10.20.53   | 10.10.20.89 | DNS      | 82     | Standard query response 0x8b2e No such name A missing-170.hatter.lab |
| 4808  | 83.610000  | 10.10.30.10   | 10.10.20.82 | HTTP     | 153    | HTTP/1.1 404 Not Found (text/plain)                                  |
| 4856  | 84.430000  | 10.10.20.53   | 10.10.20.79 | DNS      | 82     | Standard query response 0x8b4a No such name A missing-187.hatter.lab |
| 5297  | 91.740000  | 10.10.20.53   | 10.10.20.69 | DNS      | 82     | Standard query response 0x5bd5 No such name A missing-204.hatter.lab |
| 5724  | 99.050000  | 10.10.20.53   | 10.10.20.40 | DNS      | 82     | Standard query response 0xeb14 No such name A missing-221.hatter.lab |
| 11578 | 105.020000 | 10.10.30.10   | 10.10.20.65 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)                             |
| 11583 | 105.040000 | 10.10.20.65   | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0)                        |
| 11677 | 106.360000 | 10.10.20.53   | 10.10.20.92 | DNS      | 82     | Standard query response 0x2819 No such name A missing-238.hatter.lab |
| 12101 | 113.670000 | 10.10.20.53   | 10.10.20.82 | DNS      | 82     | Standard query response 0xaaa1 No such name A missing-255.hatter.lab |
| 12535 | 120.980000 | 10.10.20.53   | 10.10.20.72 | DNS      | 82     | Standard query response 0x390d No such name A missing-272.hatter.lab |
| 12737 | 124.350000 | 198.51.100.20 | 10.10.20.93 | HTTP     | 153    | HTTP/1.1 404 Not Found (text/plain)                                  |
| 12970 | 128.290000 | 10.10.20.53   | 10.10.20.62 | DNS      | 82     | Standard query response 0x25a9 No such name A missing-289.hatter.lab |
| 13374 | 135.020000 | 10.10.30.10   | 10.10.20.67 | SSHv2    | 77     | Server: Protocol (SSH-2.0-HatterLab_1.0)                             |
| 13376 | 135.040000 | 10.10.20.67   | 10.10.30.10 | SSHv2    | 82     | Client: Protocol (SSH-2.0-TrainingClient_1.0)                        |
| 13410 | 135.600000 | 10.10.20.53   | 10.10.20.95 | DNS      | 82     | Standard query response 0x267a No such name A missing-306.hatter.lab |

Figure 8. A broader view connecting all suspicious traffic.

There remains a relatively consistent pattern when putting each piece of suspicious traffic together. It generally follows the pattern of a failed HTTP connection, multiple failed DNS resolutions, and a failed SSH connection. Although, this pattern breaks when the admin login URL is identified and authorization attempts begin.

## Part 6: Security Findings

Finding One: Repeated failed SSH connections to a single target

Finding Two: HTTP connection attempts to unauthorized page

Finding Three: DNS enumeration to identify subdomains

## Part 7: Evidence

| ssh http.response.code>400 dns.flags.response==3 |           |             |             |          |        |                         |                                             |
|--------------------------------------------------|-----------|-------------|-------------|----------|--------|-------------------------|---------------------------------------------|
| Io.                                              | Time      | Source      | Destination | Protocol | Length | Info                    |                                             |
| 6                                                | 0.320000  | 10.10.20.53 | 10.10.20.25 | DNS      | 95     | Standard query response | 0x0731 A intranet.hatter.lab A 10.10.30.10  |
| 10                                               | 2.020000  | 10.10.20.53 | 10.10.20.60 | DNS      | 94     | Standard query response | 0x8f0d A updates.hatter.lab A 10.10.30.10   |
| 16                                               | 2.130000  | 10.10.30.10 | 10.10.20.60 | HTTP     | 153    | HTTP/1.1 404 Not Found  | (text/plain)                                |
| 49                                               | 3.280000  | 10.10.20.53 | 10.10.20.63 | DNS      | 94     | Standard query response | 0xa0f9 A portal.example.com A 198.51.100.20 |
| 73                                               | 4.020000  | 10.10.20.53 | 10.10.20.60 | DNS      | 80     | Standard query response | 0x390c No such name A missing-0.hatter.lab  |
| 86                                               | 4.450000  | 10.10.20.53 | 10.10.20.67 | DNS      | 94     | Standard query response | 0x798a A portal.example.com A 198.51.100.20 |
| 88                                               | 4.540000  | 10.10.20.53 | 10.10.20.66 | DNS      | 93     | Standard query response | 0x4771 A status.hatter.lab A 10.10.30.10    |
| 101                                              | 4.880000  | 10.10.20.53 | 10.10.20.74 | DNS      | 94     | Standard query response | 0xefac A updates.hatter.lab A 10.10.30.10   |
| 135                                              | 5.310000  | 10.10.20.53 | 10.10.20.81 | DNS      | 91     | Standard query response | 0x165f A www.example.org A 198.51.100.20    |
| 148                                              | 5.740000  | 10.10.20.53 | 10.10.20.88 | DNS      | 91     | Standard query response | 0xad94 A docs.hatter.lab A 10.10.30.10      |
| 152                                              | 5.800000  | 10.10.20.53 | 10.10.20.69 | DNS      | 91     | Standard query response | 0xac42 A docs.hatter.lab A 10.10.30.10      |
| 184                                              | 6.170000  | 10.10.20.53 | 10.10.20.95 | DNS      | 93     | Standard query response | 0xcd97 A status.hatter.lab A 10.10.30.10    |
| 200                                              | 6.600000  | 10.10.20.53 | 10.10.20.40 | DNS      | 95     | Standard query response | 0x7e5d A intranet.hatter.lab A 10.10.30.10  |
| 231                                              | 7.030000  | 10.10.20.53 | 10.10.20.66 | DNS      | 94     | Standard query response | 0x89dc A portal.example.com A 198.51.100.20 |
| 233                                              | 7.060000  | 10.10.20.53 | 10.10.20.72 | DNS      | 91     | Standard query response | 0x7f5b A docs.hatter.lab A 10.10.30.10      |
| 258                                              | 7.460000  | 10.10.20.53 | 10.10.20.73 | DNS      | 94     | Standard query response | 0x1850 A updates.hatter.lab A 10.10.30.10   |
| 280                                              | 7.890000  | 10.10.20.53 | 10.10.20.80 | DNS      | 91     | Standard query response | 0xdfd8 A www.example.org A 198.51.100.20    |
| 311                                              | 8.320000  | 10.10.20.53 | 10.10.20.75 | DNS      | 94     | Standard query response | 0x2553 A updates.hatter.lab A 10.10.30.10   |
| 312                                              | 8.320000  | 10.10.20.53 | 10.10.20.87 | DNS      | 91     | Standard query response | 0xe6a2 A docs.hatter.lab A 10.10.30.10      |
| 329                                              | 8.750000  | 10.10.20.53 | 10.10.20.94 | DNS      | 93     | Standard query response | 0xc1a2 A status.hatter.lab A 10.10.30.10    |
| 363                                              | 9.180000  | 10.10.20.53 | 10.10.20.15 | DNS      | 95     | Standard query response | 0x2d02 A intranet.hatter.lab A 10.10.30.10  |
| 376                                              | 9.580000  | 10.10.20.53 | 10.10.20.78 | DNS      | 91     | Standard query response | 0xc470 A docs.hatter.lab A 10.10.30.10      |
| 378                                              | 9.610000  | 10.10.20.53 | 10.10.20.65 | DNS      | 94     | Standard query response | 0x92d0 A portal.example.com A 198.51.100.20 |
| 414                                              | 10.040000 | 10.10.20.53 | 10.10.20.72 | DNS      | 94     | Standard query response | 0x68f4 A updates.hatter.lab A 10.10.30.10   |
| 427                                              | 10.470000 | 10.10.20.53 | 10.10.20.79 | DNS      | 91     | Standard query response | 0x51c7 A www.example.org A 198.51.100.20    |
| 461                                              | 10.840000 | 10.10.20.53 | 10.10.20.81 | DNS      | 94     | Standard query response | 0xbc91 A updates.hatter.lab A 10.10.30.10   |
| 464                                              | 10.900000 | 10.10.20.53 | 10.10.20.86 | DNS      | 91     | Standard query response | 0x9590 A docs.hatter.lab A 10.10.30.10      |
| 480                                              | 11.330000 | 10.10.20.53 | 10.10.20.93 | DNS      | 81     | Standard query response | 0xd9d8 No such name A missing-17.hatter.lab |
| 514                                              | 11.760000 | 10.10.20.53 | 10.10.20.25 | DNS      | 95     | Standard query response | 0x2832 A intranet.hatter.lab A 10.10.30.10  |
| 523                                              | 12.100000 | 10.10.20.53 | 10.10.20.84 | DNS      | 95     | Standard query response | 0x31f3 A intranet.hatter.lab A 10.10.30.10  |
| 541                                              | 12.100000 | 10.10.20.53 | 10.10.20.64 | DNS      | 94     | Standard query response | 0x4050 A portal.example.com A 198.51.100.20 |

Figure 9. A large volume of DNS requests are made in a rapid period of time.

In this image, we can see the querying of various subdomains associated with the hatter.lab URL from the same source IP. It happens in a very short period of time and seems to be an attempt by a malicious user to enumerate the domain in search of sensitive information. In *Figure 5*, multiple failed attempts are shown where a user is querying random but plausible URLs in hopes of finding a hidden page.

Besides the numerous failed attempts to nonexistent pages, there were also several unauthorized attempts on a hidden admin page. In *Figure 3*, failed attempts to view an admin/login page are repeated multiple times.

Finally, *Figure 2* shows multiple attempts at consistent intervals to complete an SSH connection. Although they originate from different IPs, all of these attempts fail, and combined with the consistency, it is suspected that an individual could be attempting to brute force attack the SSH credentials.

## **Part 8: Overall Security Assessment**

Overall, there seems to be an unauthorized user attempting to gain access to unauthorized information. This is evidenced by suspicious attempts to connect and access information in at least three ways: SSH connections, HTTP connections, and DNS fuzzing. For this reason, this case will be classified as a suspicious case, but it should be investigated further to prove that malicious activity occurred.

## **Part 10: Recommended Next Steps**

Recommendation One: Continue to monitor activity and connection attempts to the SSH server and admin accounts.

Recommendation Two: Investigate possible connection between the ping and arp requests to the other suspicious activity.

Recommendation Three: Increase obscurity around sensitive URLs ensuring they are not publicly exposed to reduce the risk of reconnaissance by threat actors, specifically securing the intranet.hatter.lab/admin/login page.